

The limit

$$\lim_{x \rightarrow \infty} x^2 - x.$$

is

Select the correct option

|                                  |                 |
|----------------------------------|-----------------|
| <input checked="" type="radio"/> | does not exist. |
| <input type="radio"/>            | 2.              |
| <input type="radio"/>            | 0.              |
| <input type="radio"/>            | 6.              |

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☐ -1.☐ -2.

For the function  $\frac{|x|}{x}$ , identify the false statement(s)

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Select the correct option



$$\lim_{x \rightarrow 0^+} \frac{|x|}{x} = 1$$

For the function  $\frac{|x|}{x}$ , identify the false statement(s)

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Select the correct option

|                                  |                                                       |
|----------------------------------|-------------------------------------------------------|
| <input type="radio"/>            | $\lim_{x \rightarrow 0^+} \frac{ x }{x} = 1$          |
| <input type="radio"/>            | $\lim_{x \rightarrow 0^-} \frac{ x }{x} = -1$         |
| <input type="radio"/>            | $\lim_{x \rightarrow 0} \frac{ x }{x}$ doesn't exist. |
| <input checked="" type="radio"/> | $\lim_{x \rightarrow 0} \frac{ x }{x} = 1$            |

For the function  $f(x) = x \sin \frac{1}{x}$ ,  $x \neq 0$ , which statement is true

For the function  $f(x) = x \sin \frac{1}{x}$ ,  $x \neq 0$ , which statement is true

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Select the correct option

|                                  |                                         |
|----------------------------------|-----------------------------------------|
| <input type="radio"/>            | $\lim_{x \rightarrow 1/\pi} f(x) = 0.$  |
| <input type="radio"/>            | The function is not defined at $x = 0.$ |
| <input type="radio"/>            | $\lim_{x \rightarrow 0} f(x) = 1.$      |
| <input checked="" type="radio"/> | $\lim_{x \rightarrow 0} f(x) = 0.$      |

For the function defined as

$$f(x) = cx,$$

for every  $\epsilon > 0$  the formal definition ensures

Select the correct option

|                                  |                                                                                                     |                                                      |
|----------------------------------|-----------------------------------------------------------------------------------------------------|------------------------------------------------------|
| <input type="radio"/>            | $ f(x) - cx_0  < \epsilon, \quad  x - x_0  < \delta, \text{ where } 0 < \delta < \epsilon.$         | <i>AHMAD MSC(MATH)</i><br><i>Areeb2113@gmail.com</i> |
| <input checked="" type="radio"/> | $ f(x) - cx_0  < \epsilon, \quad  x - x_0  < \delta, \text{ where } 0 < \delta < \epsilon/ c .$     |                                                      |
| <input type="radio"/>            | $ f(x) - cx_0  < \epsilon, \quad  x - x_0  < \delta, \text{ where } -\epsilon < \delta < \epsilon.$ |                                                      |
| <input type="radio"/>            | $ f(x) - cx_0  < \epsilon, \quad  x - x_0  < \delta, \text{ where } 0 < \delta < \epsilon.$         |                                                      |

The value of  $x + \lim_{x \rightarrow 0} \frac{|x|}{x} =$

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Select the correct option



$x - 1.$



$x + 1.$



$x.$



$x + 1/x.$

The limit

$$\lim_{x \rightarrow 1} \sqrt{x} - 1.$$

Select the correct option

|                                  |                 |                                                      |
|----------------------------------|-----------------|------------------------------------------------------|
| <input type="radio"/>            | 1.              | <i>AHMAD MSC(MATH)</i><br><i>Areeb2113@gmail.com</i> |
| <input checked="" type="radio"/> | 0.              |                                                      |
| <input type="radio"/>            | does not exist. |                                                      |
| <input type="radio"/>            | -1.             |                                                      |

If  $f(x) = \log x$  and  $g(x) = \frac{1}{1-x^2}$ , then  $f \circ g$  is

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Select the correct option

|                                  |                                           |
|----------------------------------|-------------------------------------------|
| <input checked="" type="radio"/> | $(f \circ g)(x) = \log \frac{1}{1-x^2}.$  |
| <input type="radio"/>            | $(f \circ g)(x) = \log \frac{x}{1-x^2}.$  |
| <input type="radio"/>            | $(f \circ g)(x) = \log \frac{1}{x^2}.$    |
| <input type="radio"/>            | $(f \circ g)(x) = \log \frac{-1}{1-x^2}.$ |



The  $\lim_{x \rightarrow 0} \frac{4 - 4 \cos x - 2 \sin^2 x}{x^4}$  is

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Select the correct option



$\frac{1}{4}$



$-\frac{1}{2}$



$\frac{1}{2}$



$2$

If  $\lim_{x \rightarrow 0} f(x) = 10$  and  $\lim_{x \rightarrow 0} g(x) = -2$  then  $\lim_{x \rightarrow 0} \left( \frac{f(x)}{g(x)} \right)$  is

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Select the correct option

|                                  |      |
|----------------------------------|------|
| <input type="radio"/>            | -10. |
| <input type="radio"/>            | 5.   |
| <input type="radio"/>            | 0.   |
| <input checked="" type="radio"/> | -5.  |

$$|f(x) - cx_0| < \epsilon, \quad |x - x_0| < \delta, \text{ where } 0 < \delta < \epsilon.$$

$$|f(x) - cx_0| < \epsilon, \quad |x - x_0| < \delta, \text{ where } 0 < \delta < \epsilon/|c|.$$

$$|f(x) - cx_0| < \epsilon, \quad |x - x_0| < \delta, \text{ where } -\epsilon < \delta < \epsilon.$$

$$|f(x) - c| < \epsilon, \quad |x - x_0| < \delta, \text{ where } 0 < \delta < \epsilon.$$

value of the limit

$$\lim_{x \rightarrow 0^+} x \log x,$$

value of the limit

$$\lim_{x \rightarrow 0^+} x \log x,$$

the correct option

0

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1

2

3

or the piecewise defined function  $f(x) = \begin{cases} \end{cases}$

$$\begin{cases} x^3, & x \leq 0, \\ x^2 \sin \frac{1}{x}, & x > 0, \end{cases}$$

Select the correct option

|                                  |    |                                                      |
|----------------------------------|----|------------------------------------------------------|
| <input checked="" type="radio"/> | 0  | <i>AHMAD MSC(MATH)</i><br><i>Areeb2113@gmail.com</i> |
| <input type="radio"/>            | 2  |                                                      |
| <input type="radio"/>            | 1  |                                                      |
| <input type="radio"/>            | -1 |                                                      |

value of  $x + \lim_{x \rightarrow 0+} \frac{1}{x} =$

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ct the correct option

☐  $x - 1.$

☒  $x + 1.$

☐  $x.$

☐  $x + 1/x.$

the value of  $x + \lim_{x \rightarrow 0^+} \frac{|x|}{x} =$

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Select the correct option

|                       |            |
|-----------------------|------------|
| <input type="radio"/> | $x - 1.$   |
| <input type="radio"/> | $x + 1.$   |
| <input type="radio"/> | $x.$       |
| <input type="radio"/> | $x + 1/x.$ |

Rolle's Theorem says, suppose that  $f$  is continuous on the closed interval  $[a, b]$  and differentiable on the open interval  $(a, b)$  and  $f(a) = f(b)$ .

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Select the correct option

|                                  |                                                                    |
|----------------------------------|--------------------------------------------------------------------|
| <input checked="" type="radio"/> | Then $f'(c) = 0$ for some $c$ in the open interval $(a, b)$ .      |
| <input type="radio"/>            | Then $f'(c) \neq 0$ for some $c$ in the open interval $(a, b)$ .   |
| <input type="radio"/>            | Then $f'(c)$ exists for some $c$ in the open interval $(a, b)$ .   |
| <input type="radio"/>            | Then $f'(c) = 0$ for some $c$ in the open interval $(a, \infty)$ . |



The function  $g(x) = x^2$  is

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Select the correct option

|                                  |                                           |
|----------------------------------|-------------------------------------------|
| <input type="radio"/>            | decreasing on $[0, \infty)$ .             |
| <input checked="" type="radio"/> | increasing on $[0, \infty)$ .             |
| <input type="radio"/>            | having negative values on $[0, \infty)$ . |
| <input type="radio"/>            | undefined on $[0, \infty)$ .              |

the derivative of the function  $x^n$  is

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Select the correct option

|                                  |                       |
|----------------------------------|-----------------------|
| <input type="radio"/>            | $x^{n-1}$ .           |
| <input checked="" type="radio"/> | $nx^{n-1}$ .          |
| <input type="radio"/>            | $\frac{n}{x^{n-1}}$ . |
| <input type="radio"/>            | $nx^n$ .              |

The function  $h(x) = -x^3$  is

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Select the correct option

|                                  |                                       |
|----------------------------------|---------------------------------------|
| <input type="radio"/>            | is undefined on $(-\infty, \infty)$ . |
| <input checked="" type="radio"/> | decreasing on $(-\infty, \infty)$ .   |
| <input type="radio"/>            | increasing on $(-\infty, \infty)$ .   |
| <input type="radio"/>            | positive on $(-\infty, \infty)$ .     |

If a function  $f$  is continuous on the closed interval then  $f$  attains its

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Select the correct option

|                                  |                                                              |
|----------------------------------|--------------------------------------------------------------|
| <input checked="" type="radio"/> | extreme values in the closed interval.                       |
| <input type="radio"/>            | extreme values at one point in the closed interval.          |
| <input type="radio"/>            | may or may not attain extreme values in the closed interval. |
| <input type="radio"/>            | derivative in the closed interval.                           |

The Mean Value Theorem says, suppose that  $f$  is differentiable on  $[a, b]$ ,  $f'(a) \neq f'(b)$ , and  $\mu$  is between  $f'(a)$  and  $f'(b)$ .

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Select the correct option

☐ Then  $f'(c) \neq \mu$  for some  $c$  in  $(a, b)$ .

☒ Then  $f'(c) = \mu$  for some  $c$  in  $(a, b)$ .

☐ Then  $f'(c) = 0$  for some  $c$  in  $(a, b)$ .

☐ Then  $f'(c)$  may be 0 or  $\mu$  for some  $c$  in  $(a, b)$ .

The inverse of the function  $f(x) = x^2$ , is

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Select the correct option

|                                  |                             |
|----------------------------------|-----------------------------|
| <input type="radio"/>            | $f^{-1}(y) = y^2$           |
| <input checked="" type="radio"/> | $f^{-1}(y) = \sqrt{y}.$     |
| <input type="radio"/>            | $f^{-1}(y) = \frac{1}{x^2}$ |
| <input type="radio"/>            | $f^{-1}(y) = \frac{1}{y^2}$ |

The inverse of the given function  $f(x) = 2x + 4, \quad 0 \leq x \leq 2$ , is

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Select the correct option



$$f^{-1}(y) = \frac{(y - 4)}{2}$$



$$f^{-1}(y) = \frac{(y - 2)}{4}$$



$$f^{-1}(y) = \frac{(x - 2)}{4}$$



$$f^{-1}(y) = \frac{(y - 4)}{4}$$

Generalized Mean Value Theorem says. If  $f$  and  $g$  are continuous on the closed interval  $[a, b]$  and differentiable on the open interval  $(a, b)$ , then

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|                                  |                                           |                                           |
|----------------------------------|-------------------------------------------|-------------------------------------------|
| <input checked="" type="radio"/> | for some $c$ in $(a, b)$ .                | $[g(b) - g(a)]f'(c) = [f(b) - f(a)]g'(c)$ |
| <input type="radio"/>            | for some $c$ in $(a, b)$ .                | $[g(b) - g(a)] = [f(b) - f(a)]g'(c)$      |
| <input type="radio"/>            | $g'(x) = 0$ for all $x$ in $(a, b)$ .     |                                           |
| <input type="radio"/>            | $g'(x) = f'(x)$ for all $x$ in $(a, b)$ . |                                           |



$\lim_{x \rightarrow 0^+} x \log x$ , has the following indeterminate form

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Select the correct option

|                                  |                          |
|----------------------------------|--------------------------|
| <input type="radio"/>            | $\infty \times \infty$ . |
| <input checked="" type="radio"/> | $(0)(\infty)$ .          |
| <input type="radio"/>            | $\frac{0}{0}$ .          |
| <input type="radio"/>            | $0^\infty$ .             |

If a function  $f$  is continuous on the closed interval then  $f$  attains its

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Select the correct option

|                                  |                                                              |
|----------------------------------|--------------------------------------------------------------|
| <input checked="" type="radio"/> | extreme values in the closed interval.                       |
| <input type="radio"/>            | extreme values at one point in the closed interval.          |
| <input type="radio"/>            | may or may not attain extreme values in the closed interval. |
| <input type="radio"/>            | derivative in the closed interval.                           |

The Mean Value Theorem says, suppose that  $f$  is differentiable on  $[a, b]$ ,  $f'(a) \neq f'(b)$ , and  $\mu$  is between  $f'(a)$  and  $f'(b)$ .

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Select the correct option

☐ Then  $f'(c) \neq \mu$  for some  $c$  in  $(a, b)$ .

☒ Then  $f'(c) = \mu$  for some  $c$  in  $(a, b)$ .

☐ Then  $f'(c) = 0$  for some  $c$  in  $(a, b)$ .

☐ Then  $f'(c)$  may be 0 or  $\mu$  for some  $c$  in  $(a, b)$ .

The function  $f(x) = \begin{cases} x, & 0 \leq x < 1, \\ 2, & 1 \leq x \leq 2, \end{cases}$  is

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Select the correct option

|                                  |                              |                |
|----------------------------------|------------------------------|----------------|
| <input checked="" type="radio"/> | is nondecreasing on          | $I = [0, 2]$ . |
| <input type="radio"/>            | is nonincreasing on          | $I = [0, 2]$ . |
| <input type="radio"/>            | is having negative values on | $I = [0, 2]$ . |
| <input type="radio"/>            | is undefined on              | $I = [0, 2]$ . |

$f : I \rightarrow \mathbb{R}$  has a derivative at  $c \in I$ , then  $f$  is

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Select the correct option

|                                  |                         |
|----------------------------------|-------------------------|
| <input checked="" type="radio"/> | continuous at $c$ .     |
| <input type="radio"/>            | discontinuous at $c$ .  |
| <input type="radio"/>            | undefined at $c$ .      |
| <input type="radio"/>            | having one sided limit. |

If a function  $f$  is continuous on the closed interval then  $f$  attains its

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Select the correct option

|                                  |                                                              |
|----------------------------------|--------------------------------------------------------------|
| <input checked="" type="radio"/> | extreme values in the closed interval.                       |
| <input type="radio"/>            | extreme values at one point in the closed interval.          |
| <input type="radio"/>            | may or may not attain extreme values in the closed interval. |
| <input type="radio"/>            | derivative in the closed interval.                           |

The value of

$$\lim_{x \rightarrow +\infty} \frac{e^x}{x^2}$$

Select the correct option



0



$-\infty$ .



$\infty$ .



3

Q. prove continuity for the given function. 2 questions

Q. second mean value theorem statement

Q using mean value theorem prove that  $x^{-1}/x < \ln x < x^{-1}$  for  $x > 1$

Q if  $F'(x)=f(x)$  prove integral  $f(x)=F(a)-F(b)$  for  $[a,b]$

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Q if  $g$  is continuous at  $x_0$  and  $f$  is continuous on  $g(x_0)$  then show that  $f \circ g$  is continuous at



## Monotone Sequences

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### 1.1 Definition:

A sequence  $\{a_n\}_{n \geq 1}$  said to be **monotonically increasing**, if  $\{a_n\}_{n \geq 1} \geq a_n \forall n \geq 1$ .

A sequence  $\{a_n\}_{n \geq 1}$  said to be **monotonically decreasing**, if  $\{a_n\}_{n \geq 1} \leq a_n \forall n \geq 1$ .

can describe now the completeness property of the real numbers.

### 1.2 Completeness property

Every monotonically increasing sequence which is bounded above is convergent.

### 1.3 Theorem:

$\{a_n\}_{n \geq 1}$  is monotonically decreasing and is bounded below, it is convergent.

of:

ows from the following facts

$\{a_n\}_{n \geq 1}$  monotonically decreasing if and only if  $\{a_n\}_{n \geq 1}$  monotonically increasing.

$\{a_n\}_{n \geq 1}$  bounded below, if and only if  $\{a_n\}_{n \geq 1}$  is bounded above.

$\{a_n\}_{n \geq 1}$  is convergent if and only if  $\{a_n\}_{n \geq 1}$  is convergent.

#### 4 Examples:

Sequence  $\{2n\}_{n \geq 1}$  is monotonically increasing and is not bounded above.

Sequence  $\{\frac{1}{n}\}_{n \geq 1}$  is monotonically decreasing and is not bounded below, say by 0.

Sequence  $\{(-1)^n\}_{n \geq 1}$  is neither monotonically increasing nor decreasing.

Let  $a_1 = 1$ ,  $a_n = \frac{3a_{n-1} + 2}{6} \quad \forall n \geq 2$ . As shown in problem 2, module 0,  $a_n \geq \frac{2}{3} \quad \forall n$

**Proof:**

Follows from the following facts

- (i)  $\{a_n\}_{n \geq 1}$  monotonically decreasing if and only if  $\{a_n\}_{n \geq 1}$  monotonically increasing.
- (ii)  $\{a_n\}_{n \geq 1}$  bounded below, if and only if  $\{a_n\}_{n \geq 1}$  is bounded above.
- (iii)  $\{a_n\}_{n \geq 1}$  is convergent if and only if  $\{a_n\}_{n \geq 1}$  is convergent.

**3.1.4 Examples:**

- (i) Sequence  $\{2n\}_{n \geq 1}$  is monotonically increasing and is not bounded above.
- (ii) Sequence  $\{\frac{1}{n}\}_{n \geq 1}$  is monotonically decreasing and is not bounded below, say by 0.
- (iii) Sequence  $\{(-1)^n\}_{n \geq 1}$  is neither monotonically increasing nor decreasing.
- (iv) Let  $a_1 = 1$ ,  $a_n = \frac{3a_{n-1} + 2}{6} \quad \forall n \geq 2$ . As shown in problem 2, module 0,  $a_n \geq \frac{2}{3} \quad \forall n$  and

$$\lim_{x \rightarrow x_0} f(x) = f(x_0).$$

- We say that  $f$  is *continuous from the left at  $x_0$*  if  $f$  is defined on an open interval  $(a, x_0)$  and

$$\lim_{x \rightarrow x_0^-} f(x) = f(x_0).$$

- We say that  $f$  is *continuous from the right at  $x_0$*  if  $f$  is defined on an open interval  $(x_0, b)$  and

$$\lim_{x \rightarrow x_0^+} f(x) = f(x_0).$$

#### Continuity (Formal):

- A function  $f$  is continuous at  $x_0$  if and only if  $f$  is defined on an open interval  $(a, b)$  containing  $x_0$  and for each  $\varepsilon > 0$  there is a  $\delta > 0$  such that

$$|f(x) - f(x_0)| < \varepsilon \quad \text{whenever} \quad |x - x_0| < \delta.$$

- A function  $f$  is continuous from the right at  $x_0$  if and only if  $f$  is defined on an interval  $[x_0, b)$  and for each  $\varepsilon > 0$  there is a  $\delta > 0$  such that

$$|f(x) - f(x_0)| < \varepsilon \quad \text{whenever} \quad x_0 \leq x < x_0 + \delta.$$



**Proof:** Consequently,

$$|F(x) - F(x')| \leq K|x - x'|, \quad a \leq x, x' \leq b.$$

**Theorem:** If  $f$  is integrable on  $[a, b]$  and  $a \leq c \leq b$ , then  $F(x) = \int_c^x f(t) dt$  is differentiable at any point  $x_0$  in  $(a, b)$  where  $f$  is continuous, with  $F'(x_0) = f(x_0)$ .

If  $f$  is continuous from the right at  $a$ , then  $F'_+(a) = f(a)$ . If  $f$  is continuous from the left at  $b$ , then  $F'_-(b) = f(b)$ .

**Proof:** Since

$$\frac{1}{x - x_0} \int_{x_0}^x f(t) dt = f(x_0).$$

We can write

$$\frac{F(x) - F(x_0)}{x - x_0} - f(x_0) = \frac{1}{x - x_0} \int_{x_0}^x [f(t) - f(x_0)] dt.$$

From this

$$\left| \frac{F(x) - F(x_0)}{x - x_0} - f(x_0) \right| \leq \frac{1}{|x - x_0|} \left| \int_{x_0}^x |f(t) - f(x_0)| dt \right|.$$

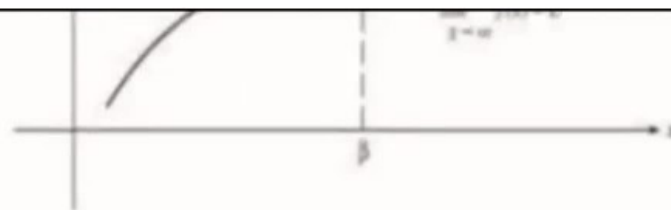


Figure 3.7: Limits at  $\pm\infty$

**Examples:** Let

$$f(x) = 1 + \frac{1}{x^2}, \quad g(x) = \frac{2|x|}{1+x}, \quad \text{and} \quad h(x) = \sin x.$$

Find limits when  $x \rightarrow \pm\infty$ .

**Solution:** We have

$$\lim_{x \rightarrow \infty} f(x) = 1,$$

since

$$|f(x) - 1| = \frac{1}{x^2} < \varepsilon \quad \text{if} \quad x > \frac{1}{\sqrt{\varepsilon}}.$$

We have

$$\lim_{x \rightarrow \infty} g(x) = 2,$$

since

$$|g(x) - 2| = \left| \frac{2x}{1+x} - 2 \right| = \frac{2}{1+x} < \frac{2}{x} < \varepsilon \quad \text{if} \quad x > \frac{2}{\varepsilon}.$$



**Theorem:** Suppose that  $f$  is continuous on the closed interval  $[a, b]$  and differentiable on the open interval  $(a, b)$ , and  $f(a) = f(b)$ . Then  $f'(c) = 0$  for some  $c$  in the open interval  $(a, b)$ .

**Proof:** Since  $f$  is continuous on  $[a, b]$ ,  $f$  attains a maximum and a minimum value on  $[a, b]$ . If these two extreme values are the same, then  $f$  is constant on  $(a, b)$ , so  $f'(x) = 0$  for all  $x$  in  $(a, b)$ .

If the extreme values differ, then at least one must be attained at some point  $c$  in the open interval  $(a, b)$ , and  $f'(c) = 0$ .

## 4.6 The Mean Value Theorem

**Theorem:** Suppose that  $f$  is differentiable on  $[a, b]$ ,  $f'(a) \neq f'(b)$ , and  $\mu$  is between  $f'(a)$  and  $f'(b)$ . Then  $f'(c) = \mu$  for some  $c$  in  $(a, b)$ .

**Proof:** Suppose first that

$$f'(a) < \mu < f'(b)$$

and define

$$g(x) = f(x) - \mu x.$$

The limit

$$\lim_{x \rightarrow 1} \sqrt{x} - 1.$$

Select the correct option

Reload Math Equations

- ☐ 1.
- ☒ 0.
- ☐ does not exist.
- ☐ -1.

Click to Save Answer & Move to Next Question



Question # 8 of 10 ( Start time: 02:13:27 PM, 28 November 2017 )

Total Marks: 1

The set of terms of  $\{s_{n_k}\}$  is contained in the set of terms of  $\{s_n\}$  implies

Select the correct option

Reload Math Equations

- |                                  |                                     |
|----------------------------------|-------------------------------------|
| <input type="radio"/>            | $\sup\{s_n\} = \sup\{s_{n_k}\}.$    |
| <input type="radio"/>            | $\sup\{s_n\} < \sup\{s_{n_k}\}.$    |
| <input checked="" type="radio"/> | $\sup\{s_n\} \geq \sup\{s_{n_k}\}.$ |
| <input type="radio"/>            | $\{s_n\}$ is convergent.            |

Question # 2 of 10 ( Start time: 10:03:30 PM, 06 February 2018 )

Total Marks: 1

The Mean Value Theorem says, suppose that  $f$  is differentiable on  $[a, b]$ ,  $f'(a) \neq f'(b)$ , and  $\mu$  is between  $f'(a)$  and  $f'(b)$ .

Select the correct option

Reload Math Equations

- ☐ Then  $f'(c) \neq \mu$  for some  $c$  in  $(a, b)$ .
- ☒ Then  $f'(c) = \mu$  for some  $c$  in  $(a, b)$ .
- ☐ Then  $f'(c) = 0$  for some  $c$  in  $(a, b)$ .
- ☐ Then  $f'(c)$  may be 0 or  $\mu$  for some  $c$  in  $(a, b)$ .

Click to Save Answer & Move to Next Question

Question # 1 of 10 ( Start time: 07:47:22 PM, 16 January 2018 )

Total Marks: 1

The limit

$$\lim_{x \rightarrow 1} x^2 - x,$$

is

Select the correct option

Reload Math Equations

does not exist.

2.

0.

6.

Click to Save Answer &amp; Move to Next Question

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## MTH621: Quiz 2

Quiz Start Time: 02:06 PM, 28 November 2017

## Question # 1 of 10 ( Start time: 02:06:09 PM, 28 November 2017 )

Total Marks: 1

Which statement(s) is(are) true about the following sequence  $s_0 = 1$  and  $s_n = 1 - e^{-s_{n-1}}$ 

Select the correct option

Reload Math Equations

 $0 < s_n \leq 1$  for all  $n$ . $s_{n+1} - s_n = -(e^{-s_n} - e^{-s_{n-1}})$ .

Sequence is convergent.



All of the above.

Type here to search

2:07 PM  
11/28/2017

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Quiz

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MC170202886: Aqsa Zunaira

Time Left 55